std::basic_fstream

Defined in header `<fstream>`

```
template<
    class CharT,
    class Traits = std::char_traits<CharT>
> class basic_fstream : public std::basic_iostream<CharT, Traits>
```

The class template `basic_fstream` implements high-level input/output operations on file based streams. It interfaces a file-based streambuffer (`std::basic_filebuf`) with the high-level interface of (`std::basic_iostream`).

A typical implementation of `std::basic_fstream` holds only one non-derived data member: an instance of `std::basic_filebuf<CharT, Traits>`.

Inheritance diagram

Several typedefs for common character types are provided:

Defined in header `<fstream>`

Type	Definition
`std::fstream`	`std::basic_fstream<char>`
`std::wfstream`	`std::basic_fstream<wchar_t>`

Member types

Member type	Definition
`char_type`	`CharT`
`traits_type`	`Traits`; the program is ill-formed if `Traits::char_type` is not `CharT`.
`int_type`	`Traits::int_type`
`pos_type`	`Traits::pos_type`
`off_type`	`Traits::off_type`
`native_handle_type`(C++26)	*implementation-defined* type that is *TriviallyCopyable* and `semiregular`

Member functions

(constructor)	constructs the file stream (public member function)
(destructor) [virtual](implicitly declared)	destructs the `basic_fstream` and the associated buffer, closes the file (virtual public member function)
`operator=` (C++11)	moves the file stream (public member function)
`swap` (C++11)	swaps two file streams (public member function)
`rdbuf`	returns the underlying raw file device object (public member function)
`native_handle` (C++26)	returns the underlying implementation-defined handle (public member function)

File operations

`is_open`	checks if the stream has an associated file (public member function)
`open`	opens a file and associates it with the stream (public member function)
`close`	closes the associated file (public member function)

Non-member functions

`std::swap``(std::basic_fstream)` (C++11)	specializes the `std::swap` algorithm (function template)

Inherited from std::basic_istream

Member functions

Formatted input

`operator>>`	extracts formatted data (public member function of `std::basic_istream<CharT,Traits>`)

Unformatted input

`get`	extracts characters (public member function of `std::basic_istream<CharT,Traits>`)
`peek`	reads the next character without extracting it (public member function of `std::basic_istream<CharT,Traits>`)

unget	unextracts a character (public member function of `std::basic_istream<CharT,Traits>`)
putback	puts a character into input stream (public member function of `std::basic_istream<CharT,Traits>`)
getline	extracts characters until the given character is found (public member function of `std::basic_istream<CharT,Traits>`)
ignore	extracts and discards characters until the given character is found (public member function of `std::basic_istream<CharT,Traits>`)
read	extracts blocks of characters (public member function of `std::basic_istream<CharT,Traits>`)
readsome	extracts already available blocks of characters (public member function of `std::basic_istream<CharT,Traits>`)
gcount	returns number of characters extracted by last unformatted input operation (public member function of `std::basic_istream<CharT,Traits>`)

Positioning

tellg	returns the input position indicator (public member function of `std::basic_istream<CharT,Traits>`)
seekg	sets the input position indicator (public member function of `std::basic_istream<CharT,Traits>`)

Miscellaneous

sync	synchronizes with the underlying storage device (public member function of `std::basic_istream<CharT,Traits>`)

Member classes

sentry	implements basic logic for preparation of the stream for input operations (public member class of `std::basic_istream<CharT,Traits>`)

Inherited from std::basic_ostream

Member functions

Formatted output

operator<<	inserts formatted data (public member function of `std::basic_ostream<CharT,Traits>`)

Unformatted output

put	inserts a character (public member function of `std::basic_ostream<CharT,Traits>`)
write	inserts blocks of characters (public member function of `std::basic_ostream<CharT,Traits>`)

Positioning

tellp	returns the output position indicator (public member function of `std::basic_ostream<CharT,Traits>`)
seekp	sets the output position indicator (public member function of `std::basic_ostream<CharT,Traits>`)

Miscellaneous

flush	synchronizes with the underlying storage device (public member function of `std::basic_ostream<CharT,Traits>`)

Member classes

sentry	implements basic logic for preparation of the stream for output operations (public member class of `std::basic_ostream<CharT,Traits>`)

Inherited from std::basic_ios

Member types

Member type	Definition
`char_type`	`CharT`
`traits_type`	`Traits`
`int_type`	`Traits::int_type`
`pos_type`	`Traits::pos_type`
`off_type`	`Traits::off_type`

Member functions

State functions

good	checks if no error has occurred i.e. I/O operations are available (public member function of `std::basic_ios<CharT,Traits>`)
eof	checks if end-of-file has been reached (public member function of `std::basic_ios<CharT,Traits>`)
fail	checks if an error has occurred (public member function of `std::basic_ios<CharT,Traits>`)
bad	checks if a non-recoverable error has occurred (public member function of `std::basic_ios<CharT,Traits>`)
operator!	checks if an error has occurred (synonym of `std::basic_ios::fail`) (public member function of `std::basic_ios<CharT,Traits>`)
operator bool	checks if no error has occurred (synonym of `!std::basic_ios::fail`) (public member function of `std::basic_ios<CharT,Traits>`)
rdstate	returns state flags (public member function of `std::basic_ios<CharT,Traits>`)
setstate	sets state flags (public member function of `std::basic_ios<CharT,Traits>`)

clear	modifies state flags (public member function of `std::basic_ios<CharT,Traits>`)

Formatting

copyfmt	copies formatting information (public member function of `std::basic_ios<CharT,Traits>`)
fill	manages the fill character (public member function of `std::basic_ios<CharT,Traits>`)

Miscellaneous

exceptions	manages exception mask (public member function of `std::basic_ios<CharT,Traits>`)
imbue	sets the locale (public member function of `std::basic_ios<CharT,Traits>`)
rdbuf	manages associated stream buffer (public member function of `std::basic_ios<CharT,Traits>`)
tie	manages tied stream (public member function of `std::basic_ios<CharT,Traits>`)
narrow	narrows characters (public member function of `std::basic_ios<CharT,Traits>`)
widen	widens characters (public member function of `std::basic_ios<CharT,Traits>`)

Inherited from std::ios_base

Member functions

Formatting

flags	manages format flags (public member function of `std::ios_base`)
setf	sets specific format flag (public member function of `std::ios_base`)
unsetf	clears specific format flag (public member function of `std::ios_base`)
precision	manages decimal precision of floating point operations (public member function of `std::ios_base`)
width	manages field width (public member function of `std::ios_base`)

Locales

imbue	sets locale (public member function of `std::ios_base`)
getloc	returns current locale (public member function of `std::ios_base`)

Internal extensible array

xalloc [static]	returns a program-wide unique integer that is safe to use as index to `pword()` and `iword()` (public static member function of `std::ios_base`)
iword	resizes the private storage if necessary and access to the `long` element at the given index (public member function of `std::ios_base`)
pword	resizes the private storage if necessary and access to the `void*` element at the given index (public member function of `std::ios_base`)

Miscellaneous

register_callback	registers event callback function (public member function of `std::ios_base`)
sync_with_stdio [static]	sets whether C++ and C I/O libraries are interoperable (public static member function of `std::ios_base`)

Member classes

failure	stream exception (public member class of `std::ios_base`)
Init	initializes standard stream objects (public member class of `std::ios_base`)

Member types and constants

Type	Explanation
openmode	stream open mode type The following constants are also defined: <table><tr><th>Constant</th><th>Explanation</th></tr><tr><td>app</td><td>seek to the end of stream before each write</td></tr><tr><td>binary</td><td>open in binary mode</td></tr><tr><td>in</td><td>open for reading</td></tr><tr><td>out</td><td>open for writing</td></tr><tr><td>trunc</td><td>discard the contents of the stream when opening</td></tr><tr><td>ate</td><td>seek to the end of stream immediately after open</td></tr><tr><td>noreplace (C++23)</td><td>open in exclusive mode</td></tr></table>(typedef)
fmtflags	formatting flags type The following constants are also defined: <table><tr><th>Constant</th><th>Explanation</th></tr><tr><td>dec</td><td>use decimal base for integer I/O: see std::dec</td></tr><tr><td>oct</td><td>use octal base for integer I/O: see std::oct</td></tr><tr><td>hex</td><td>use hexadecimal base for integer I/O: see std::hex</td></tr></table>

basefield	`dec | oct | hex`. Useful for masking operations
left	left adjustment (adds fill characters to the right): see `std::left`
right	right adjustment (adds fill characters to the left): see `std::right`
internal	internal adjustment (adds fill characters to the internal designated point): see `std::internal`
adjustfield	`left | right | internal`. Useful for masking operations
scientific	generate floating point types using scientific notation, or hex notation if combined with `fixed`: see `std::scientific`
fixed	generate floating point types using fixed notation, or hex notation if combined with `scientific`: see `std::fixed`
floatfield	`scientific | fixed`. Useful for masking operations
boolalpha	insert and extract `bool` type in alphanumeric format: see `std::boolalpha`
showbase	generate a prefix indicating the numeric base for integer output, require the currency indicator in monetary I/O: see `std::showbase`
showpoint	generate a decimal-point character unconditionally for floating-point number output: see `std::showpoint`
showpos	generate a `+` character for non-negative numeric output: see `std::showpos`
skipws	skip leading whitespace before certain input operations: see `std::skipws`
unitbuf	flush the output after each output operation: see `std::unitbuf`
uppercase	replace certain lowercase letters with their uppercase equivalents in certain output operations: see `std::uppercase`

(typedef)

iostate

state of the stream type

The following constants are also defined:

Constant	Explanation
goodbit	no error
badbit	irrecoverable stream error
failbit	input/output operation failed (formatting or extraction error)
eofbit	associated input sequence has reached end-of-file

(typedef)

seekdir

seeking direction type

The following constants are also defined:

Constant	Explanation
beg	the beginning of a stream
end	the ending of a stream
cur	the current position of stream position indicator

(typedef)

event	specifies event type (enum)
event_callback	callback function type (typedef)

Notes

Feature-test macro	Value	Std	Feature
`__cpp_lib_fstream_native_handle`	202306L	(C++26)	native handles support

Example

Run this code

```cpp
#include <fstream>
#include <iostream>
#include <string>

int main()
{
    std::string filename{"test.bin"};
    std::fstream s{filename, s.binary | s.trunc | s.in | s.out};

    if (!s.is_open())
        std::cout << "failed to open " << filename << '\n';
    else
    {
        // write
        double d{3.14};
        s.write(reinterpret_cast<char*>(&d), sizeof d); // binary output
        s << 123 << "abc";                              // text output

        // for fstream, this moves the file position pointer (both put and get)
        s.seekp(0);

        // read
        d = 2.71828;
        s.read(reinterpret_cast<char*>(&d), sizeof d); // binary input
        int n;
        std::string str;
```

```
        if (s >> n >> str)                                  // text input
            std::cout << "read back from file: " << d << ' ' << n << ' ' << str << '\n';
    }
}
```

Output:

```
read back from file: 3.14 123 abc
```

See also

`getline`	read data from an I/O stream into a string (function template)